

Introduction to Inference, Pt 3

CSU, Chico Math 351

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Binomial Framework

As we have seen, we can perform a hypothesis test for the parameter p , representing a proportion of successes out of n trials. While most of the steps are similar, the test statistic follows a slightly different formula. If X = number of successes in n trials, and $X \sim \text{Binomial}(n, p)$, then:

$$Z = \frac{X - np_0}{\sqrt{np_0(1 - p_0)}} \sim N(0, 1)$$

therefore division by n yields,

$$Z = \frac{\hat{p} - p_0}{\sqrt{\frac{p_0(1-p_0)}{n}}} \sim N(0, 1)$$

where $\hat{p} = \frac{X}{n}$

One-Sided Confidence Intervals, Reminder

When working with **proportions**, the confidence intervals won't extend to ∞ , since p can only take values between 0 and 1. Instead, one-sided confidence intervals will be bound by these limitations in place of infinity.

If testing hypothesis tests with the alternative hypothesis $H_1 : p > p_0$, use an **upper** confidence interval:

$$[\hat{p} - z_{1-\alpha}^* \cdot \sigma_{\hat{p}}, 1].$$

If testing hypothesis tests with the alternative hypothesis $H_1 : p < p_0$, use a **lower** confidence interval:

$$[0, \hat{p} + z_{1-\alpha}^* \cdot \sigma_{\hat{p}}].$$

NOTE: Since there is not a known " p_0 " for the confidence interval, use $\hat{p}$ in its place:

$$SD(\hat{p}) = \sqrt{\frac{\hat{p}(1 - \hat{p})}{n}}$$

Hypothesis Test, example

Commercial fishermen working certain parts of the Atlantic Ocean sometimes find their efforts hindered by the presence of whales. Ideally, they would like to scare away the whales without frightening the fish. One of the strategies being experimented with is to transmit underwater sounds of a killer whale. On 52 occasions that technique has been tried, it worked 24 times (that is the whales immediately left the area). Experience has shown though that 40% of all whales sighted near fishing boats leave of their own accord, probably just to get away from the boat. Let p = the proportion of whales that leave the area after hearing the underwater sounds.

Test $H_0 : p = .40$ versus $H_1 : p > .40$ at the $\alpha = .05$ level.

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Why Confidence Intervals?

The purpose of a confidence interval is to give a more informative estimator than a simple point estimator. You can achieve a certain level of confidence that your interval contains the true population mean, rather than hoping that your point estimator happens to be fairly close.

Remember, there are two parts to the margin of error in a two-sided confidence interval: **number of standard deviations** $z_{1-\alpha/2}^*$, and **size of the standard deviations** $\sigma_{\bar{X}}$.

Why Hypothesis Tests?

When you want to test a statement (H_0), a hypothesis test lets you compare your sample to the statement by determining how unlikely your test statistic is to occur under the assumption that statement is true.

We find the z-score for our test statistic $\bar{X}$ or $\hat{p}$, which represents how many standard deviations away from the mean the test statistic is.

By using `pnorm()` to find how unlikely such a test statistic is to occur, in relation to some level of confidence α , we can determine whether we reject or fail to reject the statement in question.

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Adapted from notes by Dr. Guo, Dr. Charnigo, Dr. Gray, and Dr. Roualdes.